

THERMODYNAMICS - UNIT 1 (Dr. Nishu)

Tree-Style Structured Conceptual Mindmap | Part 1: Foundations & States

1. Core Definition

- **Thermodynamics:** Branch of science dealing with transfer of energy and its effect on physical properties of substances.
- **Heat:** Movement of heat energy from one place to another / one form to another.
- **Approaches:**
 - **Macroscopic:** Classical thermodynamics, average properties (P , T , V). Uses Continuum Concept.
 - **Microscopic:** Statistical thermodynamics, molecular level tracking (positions, velocities).

2. System Classifications

- **System:** Finite quantity of matter or region in space chosen for study.
- **Surroundings:** Everything external to system.
- **Boundary:** Real/imaginary surface separating system & surroundings (fixed or moving).
- **3 Types of Systems:**
 - **Closed:** Fixed mass, energy transfer allowed, NO mass transfer (e.g., piston-cylinder with closed valve).
 - **Open:** Both mass & energy can cross boundary (e.g., turbine, compressor, pump).
 - **Isolated:** No interaction with surroundings, fixed mass & energy (e.g., Universe).

3. Properties of a System

- **Extensive:** Depend on mass/amount of substance. Additive. (e.g., Volume V , Mass m , Energy U).
- **Intensive:** Do NOT depend on mass. NOT additive. (e.g., Temperature T , Pressure P , Density ρ).
- **Specific Properties:** Extensive property per unit mass (e.g., specific volume $v = V/m$) become intensive.

4. States & Equilibrium

- **State:** Condition of a system where all properties have definite values.
- **Process:** Change of state from initial to final.
- **Thermodynamic Cycle:** Series of state changes where final state is identical to initial state ($\oint dU = 0$).
- **3 Types of Equilibrium:**
 - **Thermal:** Equality of temperature (diathermic wall).
 - **Mechanical:** Equality of forces / pressure throughout.
 - **Chemical:** Equality of chemical potential, no reaction.

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Tree-Style Structured Conceptual Mindmap | Part 2: Processes, Work, Heat & First Law

5. Process Types & Quasi-Static

- **Reversible Process:** Can be reversed without leaving any change to system or surroundings (ideal, infinitesimally slow).
- **Irreversible Process:** Cannot be reversed (e.g., friction, unrestrained expansion, mixing).
- **Quasi-Static Process:** Infinitely slow process, succession of equilibrium states. Every intermediate state is an equilibrium state.
- **Specific Processes:**
 - **Isothermal:** $T = \text{const}$ ($\Delta U = 0$).
 - **Adiabatic:** $Q = 0$ ($PV^\gamma = \text{const}$).
 - **Isobaric:** $P = \text{const}$.
 - **Isochoric:** $V = \text{const}$ ($W = 0$).

6. Work and Heat

- **Point Functions:** Properties dependent only on end states. Exact differentials (e.g., P, T, V, U).
- **Path Functions:** Depend on path followed. Inexact differentials represented by areas (e.g., Heat Q , Work W).
- **Boundary Work (Closed System):** $W = \int P dV$ (Area under P-V curve).
- **Joule's Equivalent:** $W = J \cdot Q$ where $J = 4.18 \text{ J/cal}$.

7. First Law of Thermodynamics

- **Energy Conservation:** Energy can neither be created nor destroyed, only transformed.
- **General Equation:** $Q = \Delta U + W$ or $\delta Q = dU + \delta W$.
- **Enthalpy Definition:** $H = U + PV$ (Specific Enthalpy $h = H/m$).
- **Adiabatic Derivation:** Using $C_v dT + P dV = 0$ yields $PV^\gamma = \text{const}$, where $\gamma = C_p/C_v$.

8. Limitations of 1st Law

- Considers all energy forms equivalent (hard to measure molecular energy).
- Does **NOT** consider the direction of energy transfer / conversion.
- Does not address the **efficiency** of energy conversion.
- Does not account for entropy.

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Tree-Style Structured Conceptual Mindmap | Part 3: Zeroth Law, Heat Engines, Carnot & Entropy

9. Zeroth Law & Temperature

- **Statement:** If two systems (A and C) are separately in thermal equilibrium with a third system (B), they are also in thermal equilibrium with each other ($T_A = T_B$ and $T_C = T_B \Rightarrow T_A = T_C$).
- **Basis:** Provides basis for temperature measurement using thermometric properties (e.g., length in mercury thermometer, resistance in RTDs/thermistors).

10. Heat Engines & Efficiency

- **Heat Engine:** Converts heat into work continuously between hot source (T_1) and cold sink (T_2).
- **Net Work:** $W = Q_1 - Q_2$
- **Thermal Efficiency:** $\eta = W / Q_1 = 1 - (Q_2 / Q_1)$
(Always $< 100\%$).

11. Carnot Cycle & Refrigerator

- **4 Reversible Stages:**
 1. Isothermal Expansion at T_1
 2. Adiabatic Expansion ($T_1 \rightarrow T_2$)
 3. Isothermal Compression at T_2
 4. Adiabatic Compression ($T_2 \rightarrow T_1$)
- **Carnot Efficiency:** $\eta_{Carnot} = 1 - (T_2 / T_1)$
- **Coefficient of Performance (COP):** $\beta = Q_2 / W = T_2 / (T_1 - T_2)$

12. Second Law & Entropy

- **Kelvin-Planck Statement:** Impossible to construct a device operating in a cycle that produces no effect other than work extraction and heat exchange with a single thermal reservoir.
- **Clausius Statement:** Heat cannot spontaneously flow from a colder body to a warmer body without external work.
- **Entropy (S):** Measure of molecular disorder / randomness ($dS = dQ / T$).
- **Universe Entropy:** Remains constant for reversible processes, **strictly increases** for irreversible processes.